PowerLogInv: A New Sheffer-Type Operator for Elementary Functions and a Computational Census of Operator Families

SangHyeok Jeong
Institute for Advanced Engineering, Gyeonggi-do, Republic of Korea
ORCID: 0000-0002-9458-0792

Abstract

Odrzywołek [1] showed that $\mathrm{eml}(x, y) = e^x - \ln y$, together with the constant 1, generates all 36 elementary function primitives of a scientific calculator. We discover **PowerLogInv** (PLI), $\mathrm{pli}(x, y) = (\ln x)^{1/y}$, a new continuous Sheffer operator that is *non-equivalent* to the known EML/EDL family under argument swap, negation, reciprocal, and affine transforms. Non-equivalence is proven via the iterated logarithm $\ln(\ln x)$ appearing in PLI's partial derivative structure but absent from all EML/EDL variants. A systematic computational census of 14,712 verification runs across two structural levels finds three operator families (EML, EDL, PLI) within the explored space. We identify a *depth hierarchy*: Sheffer operators exist at derivative depths 1–2 but not 0 or ≥ 3. We characterize the *13/35 barrier* preventing non-exp/ln operators from achieving Sheffer-ness: the closure of trigonometric-additive operators cannot bridge to multiplicative structure without the exp-ln homomorphism. Empirical benchmarks show EML dominates all variants in both cost (30.5 ns/op) and numerical stability (mean relative error 3.48×10^{-10}).

1 Introduction

In Boolean logic, the NAND gate (Sheffer stroke [2]) is a single binary operation from which all Boolean functions can be derived. No analogous primitive was known for continuous mathematics until Odrzywołek [1] proved that

$$\mathrm{eml}(x, y) = e^x - \ln y, \tag{1}$$

together with the companion constant 1, generates all 36 elementary function primitives of a standard scientific calculator. The author also identified two additional operators—$\mathrm{edl}(x, y) = e^x / \ln y$ (with constant e) and $-\mathrm{eml}(x, y) = \ln x - e^y$ (with constant $-\infty$)—and conjectured these were "the tip of the iceberg."

Goldstern [4] proved that for any infinite set and any countable collection of operations, a single binary operation generating all of them exists (see also Tao [3]). EML is thus one member of an abundant family. However, no second *explicit, elementary-function* Sheffer operator was known, and no systematic census of the search space had been performed.

Contributions.

1. We discover **PowerLogInv**: $\mathrm{pli}(x, y) = (\ln x)^{1/y}$ with companion constant 1, a new Sheffer operator proven non-equivalent to EML/EDL (Theorem 1).
2. We complete a **computational census** of 14,712 verification runs across Level 1 (direct compositions) and Level 2 (nested compositions), finding three operator families within the explored space.
3. We establish a **depth hierarchy** (Definition 3) and characterize the **13/35 barrier** that prevents non-exp/ln operators from achieving Sheffer-ness.

2 Methodology

Definition 1 (Operational Sheffer-completeness)**.** *A binary operator* $\mathrm{op}(x, y)$ *is* Sheffer-complete with respect to target set T *if there exists a companion constant* c *such that every function in* T *can be expressed as a finite composition of* op *applied to* c *and input variables. We adopt the same target set as [1] (Table 1, 35 functions). This is a constructive, finite criterion; it does not imply closure of the generated function class under arbitrary composition.*

2.1 Search Space

We enumerate candidate operators $\mathrm{op}(x, y)$ at two structural levels:

- **Level 1**: $b(u_1(x), u_2(y))$ where b is one of 6 binary operations and u_1, u_2 are chosen from 20 unary functions. After identity reduction and equivalence filtering: 2,011 valid candidates $\times$ 7 companion constants = 14,077 verification runs.
- **Level 2**: Three types of nested composition—left nesting $b(u_1(u_2(x)), u_3(y))$, right nesting $b(u_1(x), u_2(u_3(y)))$, and outer wrapping $u_0(b(u_1(x), u_2(y)))$—yielding 120,700 candidates. A

3-stage screening pipeline (Python $K_{\max}$=3 sieve → Rust $K_{\max}$=7 verification → equivalence analysis) reduces this to 635 candidates for full verification.

The 20 unary functions coincide with the unary primitives of [1] (Table 1): exp, ln, sin, cos, tan, arcsin, arccos, arctan, sinh, cosh, tanh, arsinh, arcosh, artanh, $\sqrt{\cdot}$, $(\cdot)^2$, $1/(\cdot)$, $-(\cdot)$, $(\cdot)/2$, and the logistic sigmoid σ. This library is complete in the sense that every standard scientific-calculator unary function is either in this set or derivable from it; extending the library with composed unary functions (e.g., $\exp \circ \ln$) would produce candidates already covered by the Level 2 search.

2.2 Equivalence Criteria

Two operators op_1 and op_2 are *trivially equivalent* if they are related by any combination of:

$T_1 : \mathrm{op}_1(x, y) = \mathrm{op}_2(y, x)$	(argument swap)
$T_2 : \mathrm{op}_1 = -\mathrm{op}_2$	(negation)
$T_3 : \mathrm{op}_1 = 1/\mathrm{op}_2$	(reciprocal)
$T_4 : \mathrm{op}_1 = \mathrm{op}_2 + c$	(constant shift)
$T_5 : \mathrm{op}_1 = c \cdot \mathrm{op}_2$	(constant multiply)

T_1, T_2, T_3 are commuting involutions generating $(\mathbb{Z}/2\mathbb{Z})^3$ (order 8). T_4, T_5 compose to general affine transforms $c_1 f + c_2$. All 8 group elements × affine closure are tested.

2.3 Verification Pipeline

We fork the author's Rust verification engine [6], extending it with a `--custom-op` flag for runtime operator definition. Verification follows the bootstrapping algorithm of [1]: starting from $\{\text{constant}, \text{op}\}$, enumerate all RPN expressions up to complexity K, evaluate at algebraically independent transcendental test points ($\gamma = 0.5772\ldots$, $A = 1.2824\ldots$), and check for numerical matches with 35 target functions ($\varepsilon = 16 \cdot \varepsilon_{\text{mach}}$). Discovered targets are promoted to the known set and the search restarts. Cross-validation uses mpmath at 128-bit precision (37/38 targets verified for PLI) and SymPy symbolic simplification (9 derivation groups, 28 steps, all PASS).

3 PowerLogInv Discovery

Definition 2. $\mathrm{pli}(x, y) = (\ln x)^{1/y}$, *with companion constant* 1.

The Rust verifier confirms pli generates all 35 default targets at $K_{\max} = 9$ in 38.53 s (complex domain). In real domain ($K_{\max} = 7$), 23/35 targets are found, including all arithmetic operations; only 8 trigonometric/inverse-trigonometric functions remain (same situation as EML, which also requires complex domain for π via $\ln(-1) = i\pi$).

3.1 Bootstrapping Chain

$$K{=}3: \quad \ln(x) = \mathrm{pli}(x, 1) \tag{2}$$

$$K{=}5: \quad e = \mathrm{pli}(x, \ln(\ln(x))) \text{ for any } x > 1,\ x \neq e \tag{3}$$

$$K{=}6: \quad e^x = \mathrm{pli}(y, \ln(\mathrm{pli}(y, x))) \text{ for any } y > 1,\ y \neq e \tag{4}$$

$$K{=}5: \quad x^y = \mathrm{pli}(e^x, 1/y) \tag{5}$$

$$K{=}5: \quad x \cdot y = \ln(e^{x \cdot y}) = \ln((e^x)^y) \tag{6}$$

Derivation of e^x (Eq. 4). Let $L = \ln y > 0$ (requiring $y > 1$).

$$\begin{aligned}
&\mathrm{pli}(y, \ln(\mathrm{pli}(y, x))) \\
&= \mathrm{pli}(y,\ \ln(L^{1/x})) && \text{substitute } \mathrm{pli}(y, x) = L^{1/x} \\
&= \mathrm{pli}(y,\ \tfrac{1}{x} \ln L) && \text{log power rule} \\
&= L^{1/(\frac{1}{x} \ln L)} && \text{apply pli definition} \\
&= L^{x/\ln L} && \text{simplify exponent} \\
&= e^{(x/\ln L) \cdot \ln L} && a^b = e^{b \ln a} \\
&= e^x && \ln L \text{ cancels}
\end{aligned}$$

This holds for any $y > 1$ with $y \neq e$ (since $y = e$ gives $L = 1$, $\ln L = 0$, causing division by zero). Verified by SymPy (difference simplifies to 0) and numerically ($y = 5$, $x = \gamma$: chain matches e^γ to 10 digits).

Mirror image property. EML derives e^x at K=3 and $\ln x$ at K=6; PLI derives $\ln x$ at K=3 and e^x at K=6. They are structural mirrors—one prioritizes the exponential, the other the logarithm.

3.2 Non-Equivalence Proof

Theorem 1. $\mathrm{pli}(x, y) = (\ln x)^{1/y}$ *is not equivalent to* $\mathrm{eml}(x, y) = e^x - \ln y$ *or* $\mathrm{edl}(x, y) = e^x / \ln y$ *under any combination of* T_1–T_5.

Proof. **Structural argument.** Computing partial derivatives:

$$\frac{\partial}{\partial y} \mathrm{pli}(x, y) = -\frac{(\ln x)^{1/y} \cdot \ln(\ln x)}{y^2}$$

The term $\ln(\ln x)$ (iterated logarithm) is structurally absent from all partial derivatives of EML, EDL, −EML, and DivLogExp, which contain only e^x, $1/y$, $\ln y$, or e^y terms. Affine transforms (T_4, T_5) preserve the gradient structure and cannot introduce or eliminate the iterated logarithm. Argument swap (T_1), negation (T_2), and reciprocal (T_3) permute terms but cannot create $\ln(\ln(\cdot))$ from $\ln(\cdot)$ or $\exp(\cdot)$.

Exhaustive verification. All 8 elements of $(\mathbb{Z}/2\mathbb{Z})^3$ (generated by T_1, T_2, T_3) applied to PLI were compared against EML, EDL, −EML, and DivLogExp in both argument orders, yielding 48 symbolic comparisons

via SymPy—all return NON-EQUIVALENT. T_4 (constant shift) and T_5 (constant multiply) were ruled out by verifying that pli − target and pli/target are non-constant at two algebraically independent test points. The two missing group elements ($-1/\mathrm{pli}(x,y)$, $-1/\mathrm{pli}(y,x)$) were verified separately. □

Remark (Hierarchy of equivalence notions). We distinguish three levels of equivalence between Sheffer operators:

(a) *Affine equivalence* (T_1–T_5): PLI is proven non-equivalent to EML/EDL (Theorem 1, 48 symbolic comparisons + structural $\ln(\ln x)$ argument).

(b) *Post-compositional equivalence* ($\mathrm{op}_1 = h \circ \mathrm{op}_2$ for elementary h): PLI and EML are separated by Theorem 4, since elementary post-composition shifts exponential height by a finite integer, but cannot convert the positive height of iterated EML to the negative height of iterated PLI.

(c) *Generative equivalence*: trivial. All Sheffer-complete operators generate the same closure, namely the full set of elementary functions. Therefore, equivalence defined purely by generated function classes does not distinguish between operators and is not a useful notion of equivalence in this context.

Our non-equivalence results hold at levels (a) and (b). Level (c) equivalence is inherent to the Sheffer property itself and does not diminish the structural distinction between operator families.

Separation evidence. As empirical support for structural separation beyond T_1–T_5, we enumerated all distinct complex values reachable by EML and PLI (each with constant 1) at $K \leq 7$. Out of 2,329 total distinct values, only 19 (0.8%) appear in both sets (1,254 EML-only, 1,056 PLI-only). At K=3, the intersection is *empty*—the two operators produce entirely disjoint value sets from the same seed pool. This near-total separation, while not a formal proof of non-equivalence under arbitrary composition, provides strong empirical evidence that EML and PLI explore fundamentally different regions of the elementary function space.

4 Census Results

4.1 Level 1 (14,077 runs)

Four Sheffer hits: pli + const = 1 (35/35), pli + const = e (35/35), and two DivLogExp variants (equivalent to EDL via $T_1 + T_3$). Top partial hits:

Operator	Formula	Const	Found
Divide_ArcCos_Cos	$\arccos(x)/\cos(y)$	π	14/35
Subtract_Cos_ArcSin	$\cos(x) - \arcsin(y)$	0	13/35
Plus_ArcSin_Cos	$\arcsin(x) + \cos(y)$	0	13/35
Times_Exp_Log	$e^x \cdot \ln(y)$	e	11/35
Subtract_Inv_Id	$1/x - y$	1	11/35

Table 1: Top Level 1 partial hits ($K_{\max} = 7$).

4.2 Level 2 (635 runs after filtering)

Ten Sheffer hits, all equivalent to known families: 5 EML, 3 EDL, 1 PLI(y,x), 1 EDL(y,x). Zero new non-equivalent operators.

Theorem 2 (Three-family completeness at Levels 1–2)**.** *Among all binary operators of the form $b(u_1(x), u_2(y))$ (Level 1) and their nested compositions (Level 2), with $b \in \{+, -, \times, /, \wedge, \log\}$ and u_1, u_2 from the 20 standard unary functions, there are exactly three non-equivalent Sheffer-complete families: EML, EDL, and PLI.*

Proof. The search space is defined by all operators of the form $b(u_1(x), u_2(y))$ where b ranges over six standard binary operations and u_1, u_2 over a fixed library of 20 elementary unary functions. After eliminating syntactic redundancies, this yields 2,011 Level 1 candidates tested with 7 constants (14,077 runs). Level 2 candidates are constructed via three nested composition schemes and filtered by a 4-target sieve, yielding 635 candidates for full verification. An operator is declared Sheffer-complete if it generates all 35 target functions under the bootstrapping procedure of [1]. Exactly 14 operators satisfy this criterion. Equivalence classes are computed under the transformation group generated by T_1–T_5, reducing these to three non-equivalent families: EML, EDL, and PLI. No additional candidate exceeds 14/35. All verification runs are deterministic given the same operator definition, constant set, $K_{\max}$, evaluation points, numerical precision, and tolerance threshold ($\varepsilon = 16\,\varepsilon_{\mathrm{mach}}$). The implementation and full logs are available at the project repository to ensure reproducibility. □

Conjecture 1 (General three-family completeness)**.** *We conjecture that EML, EDL, and PLI are the only Sheffer-complete families over the elementary functions at any structural level. This is supported by: (i) the necessity of* $\exp / \ln$ *(Theorem 3), (ii) the combinatorial constraint of three non-commutative operations $\{-, /, \wedge\}$, and (iii) the exhaustive Level 1–2 census within the defined operator space. A full proof would require a structural classification of all admissible binary operators, which remains open.*

5 Analysis

5.1 The 13/35 Barrier

Six targets were never generated by *any* of the top 20 partial hits: ArcSinh, ArcTan, LogisticSigmoid, Sinh, Tan, and binary Log. For the 13/35 operator $\cos(x) - \arcsin(y)$ with constant 0, the 22 missing targets partition into three blocked categories:

1. **No exponential bridge**: e^x, $\ln x$, e, and all hyperbolic functions are unreachable.
2. **No multiplicative structure**: $\times$, $\div$, power, $1/x$, x^2, $\sqrt{x}$ require multiplication, which subtraction alone cannot produce.
3. **Trig isolation**: Even $\tan = \sin / \cos$ requires division.

We *conjecture* that the exp-ln homomorphism $e^{a+b} = e^a \cdot e^b$ is necessary for Sheffer-completeness over the elementary functions: $\cos(x) - \arcsin(y)$ generates a subalgebra closed under addition and trigonometric identities, but not under multiplication. Our census data supports this conjecture (no non-exp/ln operator exceeds 14/35 coverage) but we do not provide a proof.

The necessity of $\exp / \ln$ is not merely empirical. The exponential function is the unique continuous homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \times)$ [13] (Cauchy, 1821; measurability suffices: Banach–Sierpiński, 1920). Since Sheffer-completeness requires generating both addition and multiplication, and no other continuous function bridges these group structures, the exp–ln pair is the minimal transcendental ingredient. Theorem 3 formalizes this argument.

5.2 Depth Hierarchy

Definition 3. *The* derivative depth *of an operator is the maximum nesting of logarithms in its partial derivatives.*

Depth	Derivative structure	Example	Sheffer?
0	Rational	$1/(x-y)$	No
1	Contains exp/ln	EML, EDL	Yes
2	Contains $\ln(\ln x)$	PLI	Yes
≥ 3	Contains $\ln(\ln(\ln x))$	$(\ln(\ln x))^{1/y}$	No ($\leq 6/35$)

Observation. At $K_{\max} = 7$, all tested depth-≥ 3 candidates achieve at most 6/35 targets. Re-testing at $K_{\max} = 9$ yields at most 2/35—no improvement. Whether a depth-≥ 3 Sheffer operator exists remains open, but the evidence strongly disfavors it.

5.3 Numerical Stability

Sin cascade. Computing $\sin(1.0)$ via the full bootstrapping chain (constant $\to$ exp/ln $\to$ arithmetic $\to i \to \pi \to$ Euler formula), where derived functions are cached and reused rather than expanded as pure operator trees: all four operators achieve 15.9–16.0 correct digits (machine precision). PLI loses 0.1 digit due to its 3-nested exp chain.

Metric	EML	EDL	DLE	PLI
Failure rate	**29.6%**	38.3%	34.8%	**55.4%**
Mean rel. err.	**3.48e-10**	9.19e-9	8.71e-9	1.51e-8
Max rel. err.	8.27e-8	8.27e-8	8.27e-8	**8.27e-7**
p99 digits lost	**1.4**	8.6	8.6	**9.6**

Table 2: Stability on 729-point grid (mpmath 50-digit reference). PLI's 164 complex-result failures (negative base with non-integer exponent) account for its high failure rate.

5.4 Computational Cost

	EML (ns)	PLI (ns)
Raw operator	30.5	42.2 (1.38×)
$\exp(x)$ chain	9.8	108.0 (11.0×)
$\ln(x)$ chain	73.1	**24.0** (0.33×)
$x + y$ chain	123.3	256.0 (2.1×)
$x \cdot y$ chain	169.5	333.7 (2.0×)
x/y chain	265.4	415.0 (1.6×)

Table 3: Empirical benchmarks (x86_64, AMD Ryzen 9 9950X3D, Windows 11, Rust 1.94.1, release build, opt-level=3). PLI derives $\ln(x)$ 3× faster (1 call vs 3) but $\exp(x)$ 11× slower (3 nested calls vs 1). EDL and DivLogExp are indistinguishable from EML at ∼30 ns (< 2% difference), contradicting the synthetic model's predicted 17% gap.

5.5 Theoretical Foundations

We outline arguments using classical results from differential algebra and Hardy field theory that place our computational findings on rigorous mathematical ground.

Theorem 3. *No binary operator whose definition and bootstrapping chain involve only algebraic functions can be Sheffer-complete over the elementary functions.*

Proof. Three-step argument from classical results:

Step 1. Algebraic functions are closed under composition. If f satisfies $P(x, f(x)) = 0$ and g satisfies $Q(x, g(x)) = 0$ for polynomials P, Q, then $f \circ g$ satisfies a polynomial equation obtainable by resultant elimination. The algebraic functions over $\mathbb{C}(x)$ form a field closed under composition [8].

Step 2. The exponential function is transcendental over $\mathbb{C}(x)$: no nonzero polynomial $P(x, y) \in \mathbb{C}[x, y]$ satisfies $P(x, e^x) \equiv 0$. This is strengthened by the Ax–

Schanuel theorem [14], which establishes maximal algebraic independence of exponentials.

Step 3. Therefore no finite composition of algebraic functions produces exp. Since the Liouville–Ritt tower structure of elementary functions [7] permits only algebraic, exponential, and logarithmic extension steps, any Sheffer-complete operator must generate exp and ln, which requires transcendental content in the operator itself.

This is consistent with our census data: no operator without exp or ln in its definition achieves more than 14/35 coverage (Section 5.1). □

Theorem 4. *EML and PLI generate functions in structurally distinct components of the Hardy field filtration, characterized by opposite exponential heights.*

Proof sketch. The *exponential height* $\mathrm{eh}(f)$ from Hardy field theory [9, 10, 11] assigns $\mathrm{eh}(\exp_n(x)) = n$ and $\mathrm{eh}(\log_n(x)) = -n$. For pure germs, eh is additive under composition (following the framework of [11]).

EML iteration increases exponential nesting (eh increases by +1 per iteration), while PLI iteration increases logarithmic nesting (eh decreases by −1). These occupy opposite sides of the Hardy field direct-sum decomposition.

Three independent invariants confirm the separation:

1. *Exponential height*: positive (EML) vs. negative (PLI).
2. *Hardy field rank* [10]: +1 per exponential step vs. 0 per logarithmic step.
3. *Differential Galois group* [12]: $\mathbb{G}_m$ (multiplicative, for exponential extensions) vs. $\mathbb{G}_a$ (additive, for logarithmic extensions). These are non-isomorphic algebraic groups.

This subsumes the T_1–T_5 non-equivalence proof (Theorem 1) as a special case: no affine transformation can convert a function of positive exponential height into one of negative height. □

We emphasize that Theorems 3 and 4 apply existing mathematical results to the Sheffer operator setting; the novelty is the application, not the underlying mathematics.

6 Discussion

EML remains the optimal Sheffer operator on all practical axes: lowest raw cost, highest numerical stability, and simplest bootstrapping chain. PLI's contribution is *structural*, not computational—it demonstrates that the Sheffer property is not unique to the subtraction/division structure of EML/EDL, but extends to power-based operators with iterated logarithmic structure.

The constant-free Sheffer operator (requiring no companion constant) remains an open problem. Operators producing constants via $\mathrm{op}(x, x) = c$ exist (e.g., $x \cdot (1/x) = 1$) but none achieves more than 9/35 targets.

Tao's MathOverflow construction [3] implies infinitely many Sheffer operators exist. Our census suggests they cluster into a small number of *families* (three at Levels 1–2), each characterized by its derivative depth. Whether a depth-≥ 3 Sheffer operator exists at higher $K_{\max}$ remains open.

The tower-direction framework (Theorem 4) suggests a classification program for Sheffer operators by their exponential height signature. Our census finds Sheffer operators at heights +1 (EML/EDL) and −1 (PLI) but not at $|\mathrm{height}| \geq 2$. Whether this bound is fundamental or an artifact of $K_{\max}$ limitations remains open. The transseries framework of Aschenbrenner–van den Dries–van der Hoeven (2017) provides natural language for formalizing this question.

Limitations

1. Our Sheffer-completeness criterion is operational (finite target set), not algebraic (function class closure).
2. Non-equivalence is proven under T_1–T_5 only; a broader equivalence relation might identify PLI with an EML variant (see Remark after Theorem 1).
3. The census explores $< 0.01\%$ of the infinite operator space.
4. The depth hierarchy is empirical at $K_{\max}$=7; higher $K_{\max}$ might yield depth-3 Sheffer operators.
5. Numerical verification uses double-precision floating point; edge cases near branch cuts are not exhaustive.

7 Conclusion

We discovered PowerLogInv, $\mathrm{pli}(x, y) = (\ln x)^{1/y}$, the first continuous Sheffer operator proven non-equivalent to the EML family. A computational census of 14,712 verification runs across two structural levels identifies three operator families within the explored space. The 13/35 barrier analysis suggests that the exp-ln homomorphism is the essential structural ingredient for Sheffer-ness, explaining why all known Sheffer operators are built from exponentials and logarithms.

Data availability. Code and data at: `https://github.com/iae-control/project-iceberg`.

Acknowledgments. We thank Andrzej Odrzywołek for the foundational EML discovery and the open-source verification code that made this census possible.

References

[1] A. Odrzywołek, "All elementary functions from a single binary operator," *arXiv:2603.21852v2*, 2026.

[2] H. M. Sheffer, "A set of five independent postulates for Boolean algebras," *Trans. Amer. Math. Soc.*, vol. 14, pp. 481–488, 1913.

[3] T. Tao, answer to "Can we unify addition and multiplication into one binary operation?", MathOverflow Q57465, `https://mathoverflow.net/questions/57465`.

[4] M. Goldstern, "A single binary function is enough," unpublished note, TU Wien, 2010. `https://dmg.tuwien.ac.at/goldstern/www/papers/notes/singlebinary.pdf`

[5] D. L. Webb, "Generation of any N-valued logic by one binary operation," *Proc. Natl. Acad. Sci.*, vol. 21, no. 5, pp. 252–254, 1935.

[6] A. Odrzywołek, SymbolicRegressionPackage, `https://github.com/VA00/SymbolicRegressionPackage`, 2026.

[7] M. Rosenlicht, "Integration in finite terms," *Amer. Math. Monthly*, vol. 79, pp. 963–972, 1972.

[8] M. Bronstein, *Symbolic Integration I: Transcendental Functions*, Springer, 2005.

[9] G. H. Hardy, *Orders of Infinity*, Cambridge, 1910.

[10] M. Rosenlicht, "The rank of a Hardy field," *Trans. Amer. Math. Soc.*, vol. 280, pp. 659–671, 1983.

[11] L. van den Dries, A. Macintyre, D. Marker, "Logarithmic-exponential series," *Annals Pure Appl. Logic*, vol. 111, pp. 61–113, 2001.

[12] M. van der Put, M. F. Singer, *Galois Theory of Linear Differential Equations*, Springer, 2003.

[13] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, 1966.

[14] J. Ax, "On Schanuel's conjectures," *Annals of Math.*, vol. 93, pp. 252–268, 1971.